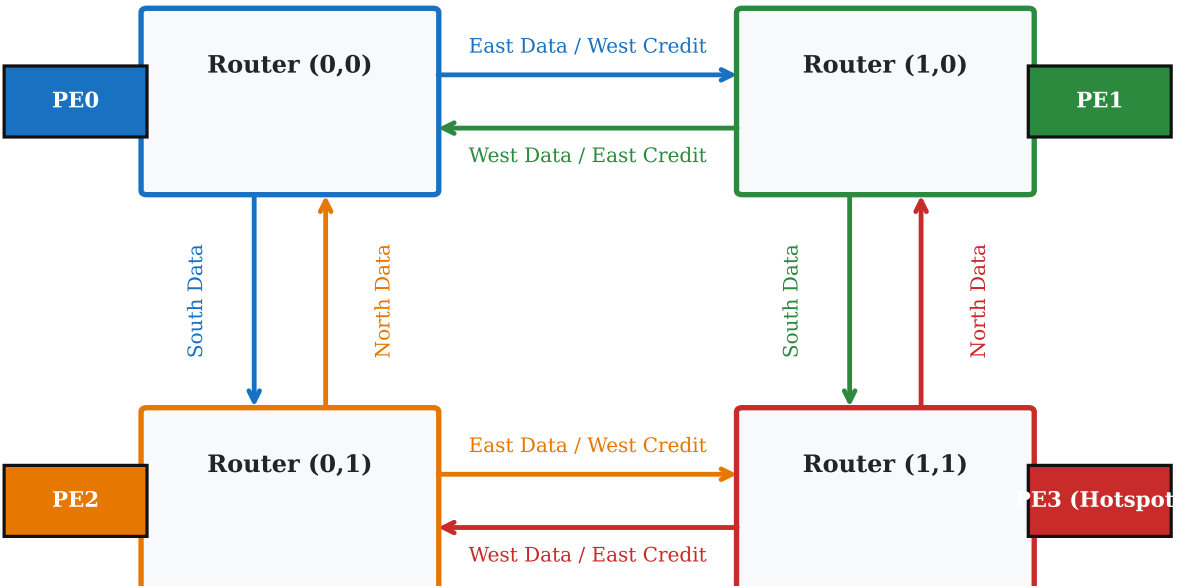2x2 2D-Mesh Interconnect Architecture